Algebra II with Trigonometry Chapter 5.1

Study guide

Saved tutoring response with MathJax rendering. Original PDF file: 4401267_alg-2trig-h-chp-5-1-note-docx.pdf

Certainly! Here's a structured study guide based on your provided document.

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1. Short Summary

This document introduces the measurement of angles in both degrees and radians, explains how to convert between these units, and covers related circular concepts such as arc length, sector area, and circular motion. It provides definitions, helpful formulas, and worked examples to support a solid understanding of foundational trigonometry and geometry concepts needed for Algebra 2/Trig or Honors classes.

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2. Core Definitions, Theorems, and Formulas

Definitions

- **Angle**: Consists of two rays sharing a common vertex; can be measured as a rotation from an initial to a terminal side.
- **Initial/Terminal Side**: The ray where measurement starts (initial) and ends (terminal).
- **Positive/Negative Angles**: Counterclockwise rotation is positive; clockwise is negative.
- **Radian**: The measure of a central angle subtended by an arc whose length equals the radius.
- **Degrees**: One full revolution = 360°, so 1° = 1/360 of a full circle.

- **Coterminal Angles**: Angles in standard position (vertex at origin, initial side on positive x-axis) that share the same terminal side.
- **Linear Speed**: Distance along a circle per unit time.
- **Angular Speed**: Angle swept per unit time (in radians).

Formulas

- **Degree–Radian Conversion**:
- Degrees to radians: $\text{radians} = \text{degrees} \times \frac{\pi}{180}$
- Radians to degrees: $\text{degrees} = \text{radians} \times \frac{180}{\pi}$

- **Arc Length** (s):
- $s = r\theta$ (where θ is in radians)

- **Area of a Sector** (A):
- $A = \frac{1}{2}r^2\theta$ (where θ is in radians)

- **Angular Speed (ω)**:
- $\omega = \frac{\theta}{t}$, where θ is in radians

- **Linear Speed (v)**:
- $v = \frac{s}{t}$ or $v = r\omega$

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3. Step-by-Step Study Guide

A. Measuring Angles

Angles can be measured in degrees or radians:

- **Degrees**: 1° = 1/360 of a full rotation.
- **Radians**: Defined as arc length divided by radius; 1 complete revolution = 2π radians.

B. Converting Degrees and Radians

- To convert degrees to radians: Multiply degrees by $\frac{\pi}{180}$
- To convert radians to degrees: Multiply radians by $\frac{180}{\pi}$

Example:

Convert 120° to radians: $120 \times \frac{\pi}{180} = \frac{2\pi}{3}$

Convert $\frac{5\pi}{6}$ radians to degrees: $\frac{5\pi}{6} \times \frac{180}{\pi} = 150^\circ$

C. Standard Position and Coterminal Angles

- **Standard position**: Angle with vertex at the origin, initial side on positive x-axis.
- **Coterminal**: Add/subtract full rotations (360° or 2π radians) to find other coterminal angles.

D. Arc Length

- For a circle with radius r and central angle θ (radians):

$$s = r\theta$$

Example:

Radius = 4, angle = $\frac{\pi}{2}$: $s = 4 \times \frac{\pi}{2} = 2\pi$

E. Area of a Sector

- Portion of a circle's area defined by central angle θ, radius r:

$$A = \frac{1}{2}r^2\theta$$

Example:

Radius = 3, angle = π: $A = \frac{1}{2} \times 3^2 \times \pi = \frac{9\pi}{2}$

F. Circular Motion

- **Angular speed (ω)**: Angle swept per unit time. $\omega = \frac{\theta}{t}$
- **Linear speed (v)**: $v = r\omega$ or $v = \frac{s}{t}$

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4. Practice Problems with Answers

1. Convert 330° to radians.

Answer: $330 \times \frac{\pi}{180} = \frac{11\pi}{6}$ radians

2. Find the length of the arc in a circle of radius 5 m subtended by a central angle of 60°.

Answer:

- First, convert 60° to radians: $60 \times \frac{\pi}{180} = \frac{\pi}{3}$
- Use $s = r\theta = 5 \times \frac{\pi}{3} = \frac{5\pi}{3}$ meters

3. In a circle of radius 8 ft, a sector has a central angle of 2 radians. What is the area of the sector?

Answer: $A = \frac{1}{2}r^2\theta = \frac{1}{2} \times 8^2 \times 2 = 64$ square feet

4. If a circle has an arc of length 18 inches that subtends an angle of $\frac{3\pi}{4}$ radians, what is the radius?

Answer: $s = r\theta \rightarrow r = \frac{s}{\theta} = \frac{18}{\frac{3\pi}{4}} = \frac{18\times4}{3\pi} = \frac{72}{3\pi} = \frac{24}{\pi}$ inches

5. If an object moves along a circle of radius 10 m and sweeps through an angle of 5π radians in 20 seconds, what are its angular and linear speeds?

Answer:

- Angular speed: $\omega = \frac{5\pi}{20} = \frac{\pi}{4}$ radians/sec
- Linear speed: $v = r\omega = 10 \times \frac{\pi}{4} = \frac{10\pi}{4} = \frac{5\pi}{2}$ meters/sec

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5. Assumptions and Ambiguities

- The document assumes all angles are in radians unless specified.
- Area of a sector and arc length formulas require the angle in radians.
- When “find the area/length” is requested, it’s implied to use the provided standard formulas and convert units as necessary.
- Example diagrams are not included here but referenced where relevant, based on the text descriptions in the PDF.

Let me know if you'd like explanations for any worked example from the document or more practice problems!